ONLINE AGRICULTURE PRODUCTS STORE

Capstone Project – Business Analysis Document

Project Sponsor	Mr. Henry / SOONY Company
Client Company	SOONY (CSR Initiative)
Development Company	APT IT Solutions
Project Manager	Mr. Vandanam
Business Analyst	Mrs Neha Bhatt (BA)
Budget	INR 2 Crores
Duration	18 Months
SDLC Model	V-Model
Technology	Java (3-Tier Architecture)

Prepared by- Neha Bhatt as a BA

Question 1 – Business Process Model (BPM)

The Business Process Model identifies the core components that define how the Online Agriculture Products Store operates.

BPM Component	Details
Goal	To provide a seamless digital platform for farmers to buy seeds, fertilizers, and pesticides from verified manufacturers, reducing procurement time and increasing agricultural productivity.
Inputs	Farmer registration data, product details from manufacturers (name, description, price, stock), payment information, delivery address, government compliance documents.
Resources	Web/Mobile Application, Java-based backend system, 3-tier architecture, Cloud servers, DB Admin (Mr. John), Network Admin (Mr. Mike), Java developers, Testers, Business Analyst.
Outputs	Confirmed orders for farmers, revenue for manufacturers, delivery tracking updates, invoices/receipts, order history reports, admin dashboards.
Activities	User Registration & Login | Product Upload by Manufacturers | Product Search & Browse by Farmers | Add to Cart | Place Order | Payment Processing | Order Dispatch & Delivery Tracking | Reviews & Ratings | Admin Monitoring.
Value to Customer	Farmers in remote areas can access quality agricultural products 24/7 without traveling long distances. Direct manufacturer communication reduces cost. Real-time tracking ensures reliability. User-friendly design ensures inclusivity.

Question 2 – SWOT Analysis

Mr. Karthik conducted the following SWOT analysis before accepting the project from SOONY Company:

SWOT Factor	Points
☑ Strengths	Strong financial backing (Rs. 2 Crores budget) | Clear project scope with defined stakeholders | Dedicated and experienced team (APT IT Solutions) | CSR initiative with social impact drives motivation | Direct farmer-manufacturer communication reduces middlemen | Existing talent pool ready for Java development
⚠ Weaknesses	Farmers in remote areas may have limited internet connectivity | Low digital literacy among the target users may hinder adoption | Application is new — no existing user base | Dependency on manufacturers to upload and maintain product listings | No prior agriculture-specific system in use (AS-IS is entirely manual)
Opportunities	Huge untapped rural farmer market across India | Government push for digital agriculture (Digital India initiative) | Potential to expand into crop selling, auctions, and agri-advisory | Integration with government schemes (e.g., PM Kisan, Soil Health Card) | Possibility of multi-language support to reach broader audience
⚡ Threats	Competition from existing platforms (e.g., AgroStar, DeHaat, Bayer Crop) | Seasonal demand fluctuations may affect revenue | Counterfeit or substandard products from unverified manufacturers | Regulatory compliance and government policy changes | Cybersecurity threats to financial transactions and user data

Question 3 – Feasibility Study

Mr. Karthik conducted a feasibility study for building the Online Agriculture Products Store using Java technology:

Feasibility Area	Assessment & Details
Hardware (HW)	Web servers for hosting application | Cloud infrastructure (AWS/Azure) for scalability | Database servers with high availability | Load balancers for traffic management | Mobile-compatible infrastructure for iOS/Android access | Estimated server cost within budget
Software (SW)	Java (Spring Boot) – Backend Development | React.js/Angular – Frontend | MySQL/PostgreSQL – Database | Apache Tomcat – Application Server | Git – Version Control | JIRA – Project Management | Selenium + JUnit – Testing | AWS/GCP – Cloud Deployment
Trained Resources	PM: Mr. Vandanam | BA: You | Sr. Java Dev: Ms. Juhi | Java Devs: Mr. Teyson, Ms. Lucie, Mr. Tucker, Mr. Bravo | DB Admin: Mr. John | NW Admin: Mr. Mike | Testers: Mr. Jason, Ms. Alekya | All resources available in APT IT Solutions talent pool
Budget	Total Budget: INR 2 Crores | Infrastructure: ~30% | Development Salaries: ~45% | Testing & QA: ~10% | Licensing & Tools: ~5% | Contingency & Miscellaneous: ~10% | Budget appears feasible for the project scope
Time Frame	Total Duration: 18 Months | RG: 1 month | RA: 2 months | Design: 2 months | Development Sprints (D1-D4): 8 months | Testing Sprints (T1-T4): 4 months | UAT: 1.5 months | Deployment: 0.5 months – Timeline is achievable

Question 4 – Gap Analysis

Gap Analysis compares the AS-IS (current/existing) process with the TO-BE (future/proposed) process:

Parameter	AS-IS (Current Process)	TO-BE (Future Process)
Product Procurement	Farmers physically travel to markets or depend on local dealers for fertilizers, seeds, and pesticides.	Farmers can browse and order products online from the comfort of their homes via web/mobile app.
Product Availability	Limited to locally available stock; farmers often face shortages, especially in remote areas.	Wide range of products from multiple certified manufacturers available 24/7.
Pricing	Prices are set by local dealers with added margins. No price comparison available.	Direct manufacturer pricing displayed. Farmers can compare products and prices easily.
Communication	No direct communication between farmers and manufacturers. Multiple middlemen involved.	Direct digital communication channel between farmers and manufacturers through the application.
Order Tracking	No order tracking. Farmers uncertain about delivery timelines.	Real-time order tracking from placement to delivery at doorstep.
Payment Mode	Cash-only transactions in most cases. No digital receipts or records.	Multiple digital payment options: UPI, Net Banking, COD, Credit/Debit cards. Digital invoices generated.
Product Information	Minimal product information available; no reviews or ratings.	Detailed product descriptions, usage guidelines, certifications, farmer reviews and ratings.
Time & Effort	Significant time and effort spent traveling to markets and waiting.	Zero travel required. Orders placed in minutes. Delivery to farm location.
Record Keeping	Manual and paper-based record keeping, prone to errors and loss.	Automated digital records for all transactions, orders, and payments.
Market Reach	Manufacturers limited to geographic reach of their distributors.	Manufacturers reach farmers across pan-India through the online platform.

Question 5 – Risk Analysis

BA Risks

Risk ID	BA Risk	Impact	Mitigation
BR01	Incomplete Requirements Gathering	High	Use multiple elicitation techniques (interviews, workshops, prototyping). Validate with all stakeholders.
BR02	Stakeholder Unavailability	Medium	Schedule regular meeting slots. Use emails/calls as backup. Document all decisions.
BR03	Scope Creep / Gold Plating	High	Maintain signed BRD. Evaluate all change requests through formal CRB process.
BR04	Miscommunication of Requirements	High	Use UML diagrams, wireframes, and use case specs to clearly communicate requirements to dev team.
BR05	Incorrect Assumptions	Medium	Document all assumptions explicitly. Validate with client regularly.

Process / Project Risks

Risk ID	Project/Process Risk	Impact	Mitigation
PR01	Internet Connectivity Issues in Rural Areas	High	Design offline mode for browsing. Optimize app for low bandwidth.
PR02	Budget Overrun	High	Quarterly audits. Fixed billing cycles. Contingency fund (10%) reserved.
PR03	Technology Risk (Java/Infra Failure)	Medium	Use proven frameworks. Conduct regular code reviews. Maintain backups.
PR04	Low User Adoption by Farmers	High	User-friendly design. Multilingual support. Training and onboarding support.
PR05	Data Security / Payment Fraud	Critical	Implement SSL, encryption, two-factor authentication, and secure payment gateways.
PR06	Delay in Regulatory Approvals	Medium	Early engagement with compliance teams. Buffer time in schedule.
PR07	Team Attrition	Medium	Knowledge transfer sessions. Documentation of all technical decisions.

Question 6 – Stakeholder Analysis (RACI Matrix)

RACI stands for: R = Responsible | A = Accountable | C = Consulted | I = Informed

Activity / Stakeholder	Mr Henry	Mr Pandu	Mr Dooku	Mr Karthik	Mr Vandanam	BA (Mrs. Neha)	Ms Juhi	Devs	Testers
Project Approval & Funding	A	C	C	R	I	C	I	I	I
Requirements Gathering	I	I	I	A	C	R	I	I	I
Business Analysis	I	I	I	C	I	R	I	I	I
System Design	I	I	I	A	C	C	R	I	I
Development	I	I	I	A	I	C	R	C	I
Testing & QA	I	I	I	A	I	C	I	R	R
UAT Coordination	A	C	C	C	I	R	I	I	C
Deployment	I	I	I	A	I	C	I	R	I
Project Monitoring	R	C	C	A	I	C	I	I	I
Change Management	A	C	C	R	I	C	I	I	I

Key Decision Makers & Influencers

Role	Person	Type
Financial Head & Approver	Mr. Pandu	Decision Maker
Project Coordinator	Mr. Dooku	Decision Maker
Sponsor & Visionary	Mr. Henry	Influencer & Decision Maker
Delivery Head	Mr. Karthik	Decision Maker
Farmer Representatives	Peter, Kevin, Ben	Influencers (Primary Stakeholders)

Question 7 – Business Case Document

Section	Details
Executive Summary	SOONY Company, under the leadership of Mr. Henry, proposes to build an Online Agriculture Products Store to solve the critical problem of agricultural input procurement faced by remote-area farmers. This CSR initiative will be developed by APT IT Solutions with a budget of INR 2 Crores over 18 months using the V-Model SDLC methodology.
Problem Statement	Remote-area farmers like Peter, Kevin, and Ben face significant challenges in procuring fertilizers, seeds, and pesticides due to limited market access, lack of information, and dependence on costly middlemen.
Proposed Solution	A web and mobile application enabling farmers to browse, compare, and purchase agricultural products directly from certified manufacturers with doorstep delivery.
Business Objectives	1. Reduce farmer procurement time by 80% | 2. Eliminate middlemen to reduce costs by 20–30% | 3. Provide 24/7 access to quality agricultural products | 4. Enable direct manufacturer-farmer communication | 5. Support 10,000+ farmers within the first year
Financial Justification	Total Investment: INR 2 Crores (CSR). Expected social impact: 10,000+ farmers benefited. Revenue model for sustainability: Commission per transaction (2–5%), premium manufacturer listings, and advertising.
Stakeholders	Mr. Henry (Sponsor), Mr. Pandu (Financial Head), Mr. Dooku (Coordinator), Mr. Karthik (Delivery Head), Farmers (Peter, Kevin, Ben), Manufacturers, APT IT Solutions Team.
Risks	Connectivity issues in rural areas, low digital literacy, scope creep, budget overruns. All identified risks have mitigation plans (see Risk Analysis section).
Timeline	18 months following V-Model: RG → RA → Design → D1/T1 → D2/T2 → D3/T3 → D4/T4 → UAT → Deployment.
Success Criteria	Application live within 18 months and budget. Minimum 1000 registered farmers in Month 1. System uptime of 99.5%. Page load time under 2 seconds.

Question 8 – Four SDLC Methodologies

Methodology	Description	When to Use
Waterfall Model	Phases are executed one after another in a strict linear order. Each phase must be completed before the next begins. Example: Waterfall Model.	When requirements are clear, fixed, and well-documented. Small projects with no expected changes.
V- Model	An extension of the Waterfall model where testing is planned alongside development.	Projects requiring high accuracy and testing, such as medical or banking software.
Spiral Model	System evolves through multiple versions. Each version is a complete system that improves over time.	High-risk projects where prototyping and risk analysis are critical.
Agile	Highly collaborative, flexible approach using short sprints (2–4 weeks). Continuous feedback from stakeholders. Example: Scrum.	When requirements change frequently. Projects needing rapid delivery and stakeholder collaboration.

Question 9 – Waterfall, RUP, Spiral and Scrum Models

Model	Key Characteristics	Best For
Waterfall	Linear sequential phases: Requirements → Design → Development → Testing → Deployment. No going back once phase is complete. Simple and easy to manage.	Well-defined projects with stable requirements.
RUP (Rational Unified Process)	Iterative framework with 4 phases: Inception, Elaboration, Construction, Transition. Use-case driven. Architecture-centric. Risk-focused.	Large enterprise projects needing structured iteration.
Spiral Model	Combines iterative development with risk analysis. Each spiral cycle: Plan → Risk Analysis → Engineering → Evaluation. High emphasis on prototyping.	High-risk, complex, long-duration projects.
Scrum (Agile)	Work divided into Sprints (2–4 weeks). Roles: Scrum Master, Product Owner, Development Team. Artifacts: Product Backlog, Sprint Backlog, Increment. Daily Stand-ups.	Dynamic projects with changing requirements and need for frequent delivery.

SME vs Project Team Debate: V-Model vs Waterfall

As a Business Analyst, I recommend the V-Model for this project. The key reasoning is:

- V-Model emphasizes testing at every stage, which is critical for a user-facing application serving low-tech-literacy farmers.
- Each development phase has a corresponding testing phase, reducing defect leakage.
- For a fixed-budget, fixed-timeline CSR project like this, structured verification at each level reduces rework costs.
- Waterfall, while simpler, lacks integrated testing discipline that V-Model provides.

Question 10 – Waterfall vs V-Model

Parameter	Waterfall Model	V-Model (Verification & Validation)
Structure	Linear sequential flow	V-shaped with parallel testing phases
Testing	Testing occurs only after development is complete	Testing planned at each development phase
Phases	Requirements → Design → Development → Testing → Deployment	Requirements ↔ UAT | System Design ↔ System Test | HLD ↔ Integration Test | LLD ↔ Unit Test | Coding
Error Detection	Errors discovered late (expensive to fix)	Errors detected early through verification
Documentation	Documentation-heavy but testing docs created later	Testing documents created alongside development docs
Flexibility	Very rigid; difficult to accommodate changes	Rigid but with built-in quality checks
Cost of Defects	High – defects found late cost more to fix	Lower – defects caught early at each level
Best For	Simple projects with clear requirements	Projects requiring high quality and early defect detection

Question 11 – Justification for Choosing V-Model

As a Business Analyst on this project, I recommend V-Model for the following reasons:

1. User-Friendliness is Critical: Since farmers are the primary users with varying digital literacy, the application must be defect-free and intuitive. V-Model's rigorous testing at each stage ensures high quality.
2. Fixed Budget & Timeline: With INR 2 Crores and 18 months, there is no room for expensive late-stage defect fixing. V-Model catches errors early, saving cost.
3. Well-Defined Requirements: Requirements are gathered from clear stakeholders (farmers, manufacturers, committee). V-Model is ideal when requirements are relatively stable.
4. Regulatory Compliance: Agriculture platforms in India need to comply with government regulations. V-Model's validation phases ensure compliance is verified at each stage.
5. High-Stakes Project (CSR): Being a CSR initiative of SOONY under Mr. Henry's vision, delivering a high-quality, reliable product is paramount. V-Model's structured verification provides the necessary quality assurance.
6. Parallel Testing Preparation: Testing teams (Mr. Jason, Ms. Alekya) and BA can prepare test cases during the design phase, optimizing 18-month timeline.

Question 12 – Gantt Chart (V-Model)

The following Gantt chart illustrates the project timeline across 18 months for the Online Agriculture Products Store using the V-Model. Resources are mapped to each phase.

PROJECT GANTT CHART – 18 MONTHS

Phase / Activity	M 1	M 2	M 3	M 4	M 5	M 6	M 7	M 8	M 9	M 10	M 11	M 12	M 13	M 14	M 15	M 16	M 17	M 18	Resources
RG (Requirements Gathering)																			PM, BA, Stakeholders
RA (Requirements Analysis)																			BA, PM
Design																			BA, Java Dev, DB Admin
D1 (Development Sprint 1)																			Java Dev, DB Admin
T1 (Testing Sprint 1)																			Testers, BA
D2 (Development Sprint 2)																			Java Dev, DB Admin
T2 (Testing Sprint 2)																			Testers, BA
D3 (Development Sprint 3)																			Java Dev, NW Admin
T3 (Testing Sprint 3)																			Testers, BA
D4 (Development Sprint 4)																			Java Dev, DB Admin

Phase / Activity	M 1	M 2	M 3	M 4	M 5	M 6	M 7	M 8	M 9	M 10	M 11	M 12	M 13	M 14	M 15	M 16	M 17	M 18	Resources
T4 (Testing Sprint 4)																			Testers, BA
UAT (User Acceptance Testing)																			BA, PM, Stakeholders
Deployment & Implementation																			All Resources

Resource Allocation Summary

Phase	Primary Resources	Duration	Months
RG – Requirements Gathering	PM (Mr. Vandanam), BA (Neha), Stakeholders (Peter, Kevin, Ben)	2 weeks	M1
RA – Requirements Analysis	BA (Neha), PM	4 weeks	M1–M2
Design	BA, Ms. Juhi, DB Admin (Mr. John)	4 weeks	M2–M4
D1 – Development Sprint 1	All Java Devs, DB Admin	6 weeks	M4–M6
T1 – Testing Sprint 1	Mr. Jason, Ms. Alekya, BA	4 weeks	M6–M7
D2 – Development Sprint 2	All Java Devs, DB Admin	6 weeks	M7–M9
T2 – Testing Sprint 2	Mr. Jason, Ms. Alekya, BA	4 weeks	M9–M10
D3 – Development Sprint 3	All Java Devs, NW Admin (Mr. Mike)	6 weeks	M10–M12
T3 – Testing Sprint 3	Mr. Jason, Ms. Alekya, BA	4 weeks	M12–M13
D4 – Development Sprint 4	All Java Devs, DB Admin	4 weeks	M13–M15
T4 – Testing Sprint 4	Mr. Jason, Ms. Alekya, BA	4 weeks	M15–M16
UAT	BA, PM, Peter, Kevin, Ben, Committee	6 weeks	M16–M17
Deployment & Implementation	All Resources	4 weeks	M17–M18

Question 13 – Fixed Bid vs Billing Model

Parameter	Fixed Bid Model	Billing Model (Time & Material)
Definition	A pre-agreed fixed price for the entire project scope.	Client pays based on actual hours/resources used, typically fortnightly.
Risk Bearer	Vendor (APT IT Solutions) bears cost overrun risk.	Client (SOONY) bears cost overrun risk.
Scope Changes	Changes are difficult and may require additional negotiations.	Changes are easily accommodated in the next billing cycle.
Transparency	Less transparent – client pays lump sum.	Highly transparent – client sees timesheets and approves payments.
Best For	Well-defined, stable-scope projects.	Projects with evolving requirements or uncertain scope.
Chosen Model	—	☑ SOONY Committee chose Billing Model with fortnightly timesheets, 3-day verification cycle, and Quarterly Audits.

Questions 14–20 – Sample BA Timesheets

Below are sample weekly timesheets for a Business Analyst across all SDLC stages of the V-Model.

RG – Requirements Gathering Timesheet

Day	Activities (8 Hours/Day)
Monday	Stakeholder interviews with Peter, Kevin, Ben – 8hrs
Tuesday	Prepare interview notes and question bank – 8hrs
Wednesday	Workshop with Mr. Henry's committee – 8hrs
Thursday	Document elicitation outputs – 8hrs
Friday	Review with PM, circulate for sign-off – 8hrs

RA – Requirements Analysis Timesheet

Day	Activities (8 Hours/Day)
Monday	Analyze gathered requirements, identify gaps – 8hrs
Tuesday	Prepare Business Requirements Document (BRD) – 8hrs
Wednesday	Review BRD with stakeholders – 8hrs
Thursday	Update BRD based on feedback – 8hrs
Friday	Prepare Use Case Diagram and Use Case Specs – 8hrs

Design Timesheet

Day	Activities (8 Hours/Day)
Monday	Create Activity Diagrams and DFDs – 8hrs
Tuesday	Design wireframes/screen mockups – 8hrs
Wednesday	Review wireframes with dev team – 8hrs
Thursday	Finalize Functional Requirements Specification – 8hrs
Friday	Update RTM and get design sign-off – 8hrs

Development Timesheet

Day	Activities (8 Hours/Day)
Monday	Attend daily stand-ups with dev team – 1hr | Clarify FR queries from developers – 4hrs | Update RTM – 3hrs
Tuesday	Review developed modules against requirements – 8hrs
Wednesday	Prepare test scenarios for Test team – 8hrs
Thursday	Conduct sprint review with PM – 2hrs | Document clarifications – 6hrs
Friday	Update project status for committee – 2hrs | Review next sprint backlog – 6hrs

Testing Timesheet

Day	Activities (8 Hours/Day)
Monday	Review Test Cases prepared by testing team – 8hrs
Tuesday	Map test cases to requirements in RTM – 8hrs
Wednesday	Participate in defect triage meetings – 4hrs | Document defects – 4hrs
Thursday	Retest resolved defects – 4hrs | Update RTM status – 4hrs
Friday	Prepare testing summary report for PM – 8hrs

UAT Timesheet

Day	Activities (8 Hours/Day)
Monday	Prepare UAT Test Plan and schedule – 8hrs
Tuesday	Conduct UAT onboarding session with Peter, Kevin, Ben – 8hrs
Wednesday	Support UAT execution, capture feedback – 8hrs
Thursday	Document UAT defects, coordinate fixes – 8hrs
Friday	Obtain UAT sign-off, prepare Client Acceptance Form – 8hrs

Deployment & Implementation Timesheet

Day	Activities (8 Hours/Day)
Monday	Coordinate with NW Admin (Mr. Mike) for server setup – 8hrs
Tuesday	Validate production deployment checklist – 8hrs
Wednesday	Support go-live; monitor for issues – 8hrs
Thursday	Prepare user manuals and training materials – 8hrs
Friday	Conduct Project Closure Documentation – 8hrs

Question 22 – BA Approach Strategy

Before project kick-start, the following BA Approach Strategy was presented to Mr. Karthik and the Committee:

1. Elicitation Techniques to Apply

- Interviews – One-on-one sessions with Peter, Kevin, Ben, Mr. Henry, and manufacturers
- Workshops/Brainstorming – Group sessions with the committee and development team
- Document Analysis – Review existing agriculture portal documentation and compliance docs
- Prototyping – Build wireframes/screen mockups for farmer and manufacturer workflows
- Use Case Specifications – Document detailed use cases for all identified scenarios

2. Stakeholder Analysis

- RACI Matrix – Identify who is Responsible, Accountable, Consulted, and Informed for each activity
- ILS (Interest, Level of influence, Support) – Map stakeholders by influence and interest to prioritize communication

3. Documents to Produce

- BPM | SWOT | Gap Analysis | Risk Register | Business Case | RACI Matrix
- Business Requirements Document (BRD) | Functional Requirements Specification (FRS)
- Use Case Diagram | Use Case Specifications | Activity Diagrams | Wireframes | DFD | ER Diagram
- RTM (Requirements Traceability Matrix) | Test Cases | UAT Plan | Project Closure Document

4. Sign-Off & Approval Process

- Draft document → Internal review with PM (Mr. Vandanam)
- Revised draft → Share with Mr. Karthik (Delivery Head) for approval
- Approved version → Submit to SOONY Committee (Mr. Henry, Mr. Pandu, Mr. Dooku) for client sign-off
- Signed BRD stored in project repository; shared with all stakeholders

5. Communication Channels

- Email – Formal communication for document sharing, decisions, and approvals
- Video Calls / Meetings – Bi-weekly committee meetings, daily team stand-ups
- JIRA – Task tracking, sprint management, defect tracking
- Confluence/SharePoint – Document repository and collaboration
- WhatsApp/Slack – Informal quick communication within the team

6. Change Request Management

- All change requests raised through formal Change Request Form
- Assessed for impact on scope, cost, and timeline by BA and PM
- Change Request Board (CRB) – Mr. Karthik + Committee – approves/rejects requests
- Approved changes updated in BRD, FRS, and RTM; communicated to dev team

7. Stakeholder Progress Updates

- Weekly status report emailed to Mr. Karthik and Mr. Vandanam
- Fortnightly timesheet submission to Committee per billing model agreement
- Quarterly Audit presentations to SOONY Committee .

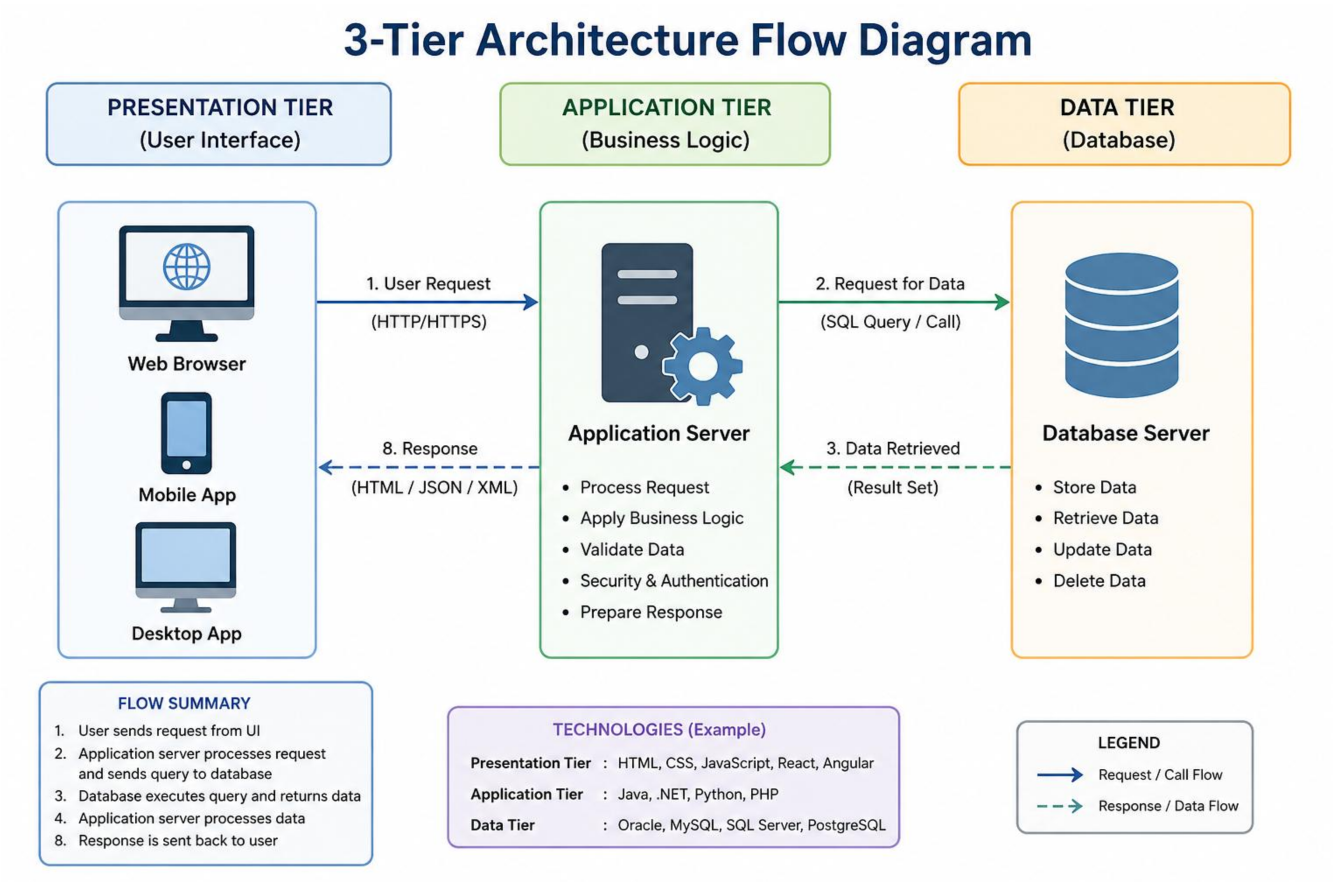
8. UAT Sign-Off Process

- BA prepares UAT Test Plan and scripts; schedules UAT sessions with Peter, Kevin, Ben
- Execute UAT, document all findings in UAT Defect Log
- On successful UAT, prepare Client Project Acceptance Form (CPAF)
- CPAF signed by Mr. Henry / Committee → Formal project handover

Question 23 – 3-Tier Architecture

The technical team has decided to use 3-tier architecture for the Online Agriculture Products Store. This separates the application into three layers:

Tier	Layer Name	Components	Technology Used
Tier 1	Presentation Layer (UI/Frontend)	Web browser interface, Mobile app screens, HTML pages, CSS styling, JavaScript interactions	React.js / Angular, HTML5, CSS3, JavaScript
Tier 2	Business Logic Layer (Application)	Application server processing, Business rules engine, Authentication & Authorization, API services, Order processing logic	Java (Spring Boot), REST APIs, Tomcat Server
Tier 3	Data Layer (Database)	Database storage, Data retrieval and storage, Query processing, Transaction management	MySQL / PostgreSQL, JDBC, Hibernate ORM



Question 24 – BA Approach Strategy for Framing Questions

Before framing any question to ask a stakeholder, a Business Analyst must consider:

Framework	Application to This Project
5W 1H	Who: Who are the users? Who approves? | What: What features are needed? | When: When is it needed? | Where: Where will it be used? | Why: Why is this required? | How: How will it work technically?
SMART	Specific: Ask clear, unambiguous questions | Measurable: Can the answer be quantified? | Achievable: Is the requirement feasible? | Relevant: Does it align with project goals? | Time-bound: When must this be delivered?
RACI	Confirm who is Responsible for each requirement, who Approves it, who should be Consulted, and who just needs to be Informed before framing questions.
3-Tier Architecture	Frame questions considering Presentation Layer (How should the UI look?), Business Layer (What logic applies?), and Data Layer (What data needs to be stored?).
Use Cases	Frame questions around each use case: Who triggers it? What is the input? What is the expected output? What are alternate flows?
Activity Diagrams	Map the process flow: What happens first? What are the decision points? What are parallel activities?
Wireframes	Use screen mockups to visually ask: Is this the correct layout? Does this flow match your expectation?

Question 25 – Elicitation Techniques (BDRFOWJIPQU)

Letter	Technique	Description
B	Brainstorming	Group sessions to generate a large number of ideas about requirements in a short time.
D	Document Analysis	Reviewing existing documents, manuals, and system specifications to identify requirements.
R	Requirements Workshop	Structured group session with key stakeholders to rapidly define and prioritize requirements.
F	Focus Groups	Gathering a group of potential users to discuss their needs and expectations.
O	Observation	Watching how users perform their tasks in their natural environment (shadowing).
W	Whiteboard/Workshop	Visual collaborative sessions using boards or tools to map out processes and requirements.
J	JAD (Joint Application Development)	Collaborative workshops with developers and stakeholders to design the system together.
I	Interview	One-on-one or group structured conversations to gather detailed requirements.
P	Prototyping	Building a working model or wireframe to validate requirements visually.
Q	Questionnaire/Survey	Structured forms sent to multiple stakeholders to gather wide-scope input efficiently.
U	Use Case Specification	Documenting detailed scenarios of how users interact with the system.

Question 26 – Elicitation Techniques for This Project

Technique	Application & Justification
☑ Prototyping	Farmers and committee members are non-technical. Creating wireframes and screen mockups allows them to visualize and validate the product early. Used for Home Page, Registration, Product Listing, Cart, and Order Tracking screens.
☑ Use Case Specifications	For every major workflow (Register, Search, Order, Upload Product, Track Order), detailed Use Case Specs ensure developers understand exactly what to build. Critical for this project given 3-tier architecture.
☑ Document Analysis	Analyzing existing agriculture product brochures, manufacturer catalogs, government compliance documents (e.g., fertilizer regulations), and similar e-commerce platforms (AgroStar) to extract implicit requirements.
☑ Brainstorming	Workshops with Peter, Kevin, Ben, Mr. Henry's committee, and development team to generate ideas, identify missing requirements, and resolve design conflicts collaboratively.

Question 27 – 10 Business Requirements

Req ID	Requirement Name	Description	Priority
BR001	Farmer Registration	Farmers should be able to register with the platform using mobile number, email, and location details.	10
BR002	Manufacturer Onboarding	Manufacturers (fertilizer, seed, pesticide companies) should be able to register and get verified by admin.	10
BR003	Product Catalog Display	The system should display a comprehensive catalog of agricultural products with descriptions, prices, and availability.	9
BR004	Product Search & Filter	Farmers should be able to search and filter products by category, brand, price range, and rating.	8
BR005	Shopping Cart & Checkout	Farmers should be able to add products to cart, review order summary, and proceed to checkout.	9
BR006	Multiple Payment Options	The system should support UPI, Net Banking, Credit/Debit Cards, and Cash on Delivery.	8
BR007	Order Tracking	Farmers should receive real-time order status updates from placement to doorstep delivery.	9
BR008	Manufacturer Product Upload	Manufacturers should be able to upload, edit, and manage their product listings with images and certifications.	10
BR009	Ratings & Reviews	Farmers should be able to rate and review products after purchase to help other farmers make informed decisions.	7
BR010	Admin Dashboard	Admin should have a dashboard to manage users, manufacturers, products, orders, and generate reports.	8

Question 28 – Assumptions

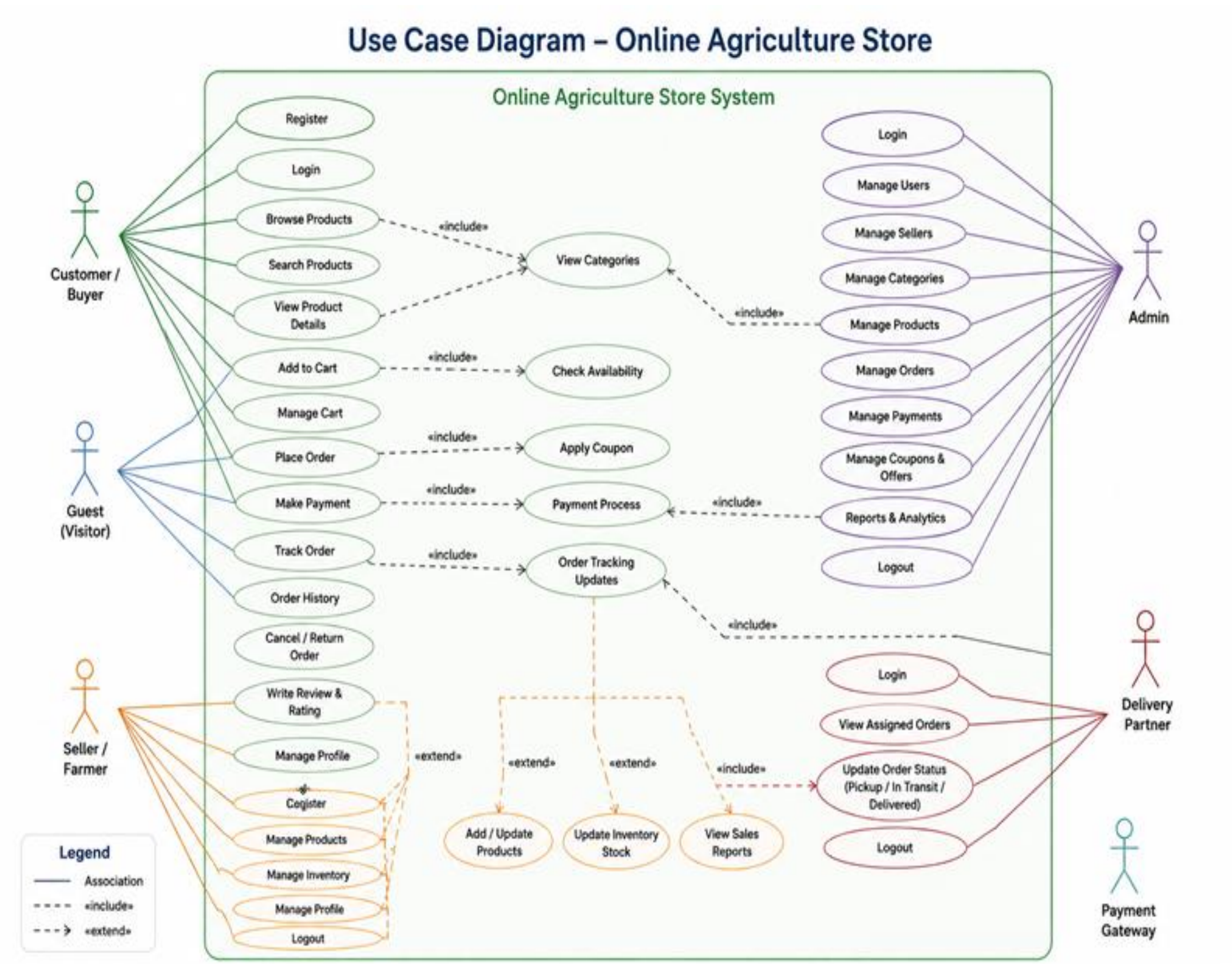
1. All farmers have access to a smartphone or basic internet-enabled mobile device.
2. Manufacturers are willing to list their products digitally and maintain updated product information.
3. The application will initially launch in English; regional language support can be added in a later phase.
4. All payments will be processed through a certified third-party payment gateway (e.g., Razorpay, PayU).
5. Delivery will be handled by third-party logistics partners; the application will integrate with their tracking APIs.
6. The Government of India's regulations on online sale of agricultural products will be complied with.
7. Internet connectivity, while limited in some rural areas, is available via mobile data (2G/3G minimum) for order placement.
8. The committee (Mr. Henry, Mr. Pandu, Mr. Dooku) will be available for bi-weekly reviews and timely approvals.
9. SOONY will ensure timely fund release per the fortnightly billing model agreement.
10. The scope excludes the development of a physical logistics network; the platform connects to existing courier services.

Question 29 – Requirements Priority Table

Req ID	Req Name	Req Description	Priority (1–10)
BR001	Farmer Search for Products	Farmers should be able to search for available products in fertilizers, seeds, pesticides	8
BR002	Manufacturers Upload Products	Manufacturers should be able to upload and display their products in the application	8
FR001	Farmer Registration	Farmers should be able to register with the application	10
FR002	Manufacturer Registration & Verification	Manufacturers register and are verified by admin	10
FR003	Product Catalog Browsing	Display full catalog with images, descriptions, and prices	9
FR004	Shopping Cart & Checkout	Cart management, order summary, and checkout flow	9
FR005	Order Placement	Farmers place orders with delivery address	9
FR006	Payment Processing	Support UPI, Net Banking, COD, Cards	8
FR007	Order Tracking	Real-time order status from placement to delivery	9
FR008	Admin Dashboard	User management, product approvals, reporting	8
NFR001	Page Loading Time	Each page loads within 2 seconds	9
NFR002	Security	SSL, 2FA, encrypted payments	10

Question 30 – Use Case Diagram

Use Case Diagram – Online Agriculture Products Store



Legend: Farmer initiates product search, ordering, and tracking workflows. Manufacturer manages product listings. Admin oversees platform management, user approvals, and reporting.

Extended Use Case List

Use Case ID	Use Case Name	Actor(s)	Description
UC001	Register	Customer, Seller	New users create accounts on the platform.
UC002	Login / Logout	Customer, delivery, Seller, Admin	Authenticated access to the system.
UC003	Search Products	Customer, Guest	Search and filter agricultural products by category, price, brand.
UC004	View Product Details	Customer, Guest	View full product information, images, reviews.
UC005	Add to Cart	Customer, Guest	Add selected products to shopping cart.
UC006	Check Availability	System	Always triggered when placing an order — verifies stock is available before confirming.
UC007	Apply Coupon	System	Validates and applies any discount coupon code at checkout.
UC008	Place Order	Customer, Guest	Finalize cart items and submit order with delivery address.
UC009	Make Payment	Customer, Guest	Complete payment via UPI/Net Banking/COD/Card.
UC010	Payment Process	Payment Gateway	External gateway handles the actual transaction (card, UPI, net banking, etc.).
UC011	Track Order	Customer, Guest	View real-time order tracking status.
UC012	Rate & Review Product	Customer, Guest	Submit product ratings and reviews post-delivery.
UC013	Upload Product	Customer, Guest	Add new product listing with details and images.
UC014	Manage Product Catalog	Manufacturer	Edit, update, or remove product listings.
UC015	View Orders Received	Manufacturer , delivery partner	See orders placed by farmers for their products.
UC016	Manage Users	Admin	Approve/reject farmer and manufacturer registrations.
UC017	Generate Reports	Admin	Generate sales, user activity, and inventory reports.
UC018	Manage Payments & Taxes	Admin	Oversee payment reconciliation and tax compliance.
UC019	View Assigned Orders	Delivery	See the list of orders assigned to this delivery partner for fulfilment.

Question 32 – Activity Diagram

Activity Diagram: Farmer Places an Order

ACTIVITY DIAGRAM – Farmer Order Placement Flow

```
                      ●    START
                      │
                      ▼
          ┌──────────────────────┐
          │  Farmer Visits App   │
          └──────────┬───────────┘
                      │
                      ▼
          ┌──────────────────────┐
          │   Farmer Logged In?  │◄─────────┐
          └──────────┬───────────┘          │
               NO     │    YES               │
                ┌─────┘     │                │
                ▼           ▼                │
          [Login/Rgstr] [Browse Products]    │
                │           │                │
                └─────┬─────┘                │
                      ▼                      │
          ┌──────────────────────┐          │
          │  Search & Filter     │          │
          └──────────┬───────────┘          │
                      ▼                      │
          ┌──────────────────────┐          │
          │  View Product Detail │          │
          └──────────┬───────────┘          │
                      ▼                      │
          ┌──────────────────────┐          │
          │   Add to Cart        │          │
          └──────────┬───────────┘          │
                      ▼                      │
          ┌──────────────────────┐          │
          │Continue Shopping?    ├──YES─────┘
          └──────────┬───────────┘
                 NO   │
                      ▼
          ┌──────────────────────┐
          │  Review Cart         │
          └──────────┬───────────┘
                      ▼
          ┌──────────────────────┐
          │  Enter Delivery Addr │
          └──────────┬───────────┘
                      ▼
          ┌──────────────────────┐
          │  Select Payment Mode │
          └──────────┬───────────┘
```

```
                     ▼
           ┌──────────────────────┐
           │  Payment Successful? │
           └──────┬───────┬───────┘
              YES │       │NO
                  ▼       ▼
         [Order Confirm] [Retry Payment]
                  │
                  ▼
           ┌──────────────────────┐
           │  Order Confirmation  │
           │  Email/SMS Sent      │
           └──────────┬───────────┘
                     ▼
           ┌──────────────────────┐
           │  Order Tracked until │
           │  Delivered           │
           └──────────┬───────────┘
                     ▼
                     ◉  END
```

Question 33 – Functional & Non-Functional Requirements

Functional Requirements

Req ID	Req Name	Req Description	Priority
FR0001	Farmer Registration	Farmers should be able to register with name, mobile, email, address, and Aadhaar.	8
FR0002	Farmer Search for Products	Farmers should be able to search for available products in fertilizers, seeds, pesticides.	8
FR0003	Product Filter	Farmers should be able to filter products by category, price range, brand, and availability.	7
FR0004	Product Detail View	Farmers should see full product details including images, descriptions, certifications, and reviews.	8
FR0005	Shopping Cart	Farmers should be able to add, update quantity, and remove items from cart.	9
FR0006	Order Placement	Farmers should be able to place orders with delivery address and payment selection.	9
FR0007	Payment Gateway Integration	System should support UPI, Net Banking, Credit/Debit Card, and Cash on Delivery.	8
FR0008	Order Confirmation	Farmers should receive email and SMS confirmation upon successful order placement.	7
FR0009	Order Tracking	Farmers should track real-time order status from placement to delivery.	8
FR0010	Order History	Farmers should view past order history with invoice download option.	6
FR0011	Product Rating & Review	Farmers should rate (1-5 stars) and write reviews for products after delivery.	6
FR0012	Manufacturer Registration	Manufacturers should register with company details and get admin verified.	9
FR0013	Product Upload by Manufacturer	Manufacturers should upload products with name, category, description, price, stock, and images.	9
FR0014	Product Management	Manufacturers should edit, update stock, or deactivate product listings.	8
FR0015	Manufacturer Order Management	Manufacturers should view and update status of orders received.	8
FR0016	Admin User Management	Admin should approve/reject/deactivate farmer and manufacturer accounts.	8
FR0017	Admin Product Approval	Admin should review and approve manufacturer product listings before display.	8
FR0018	Admin Reporting	Admin should generate reports on sales, users, and product performance.	7
FR0019	Search Auto-Suggest	System should show auto-suggestions when farmers type in the search box.	5

Req ID	Req Name	Req Description	Priority
FR0020	Wishlist / Save for Later	Farmers should be able to save products to wishlist for future purchase.	5

Non-Functional Requirements

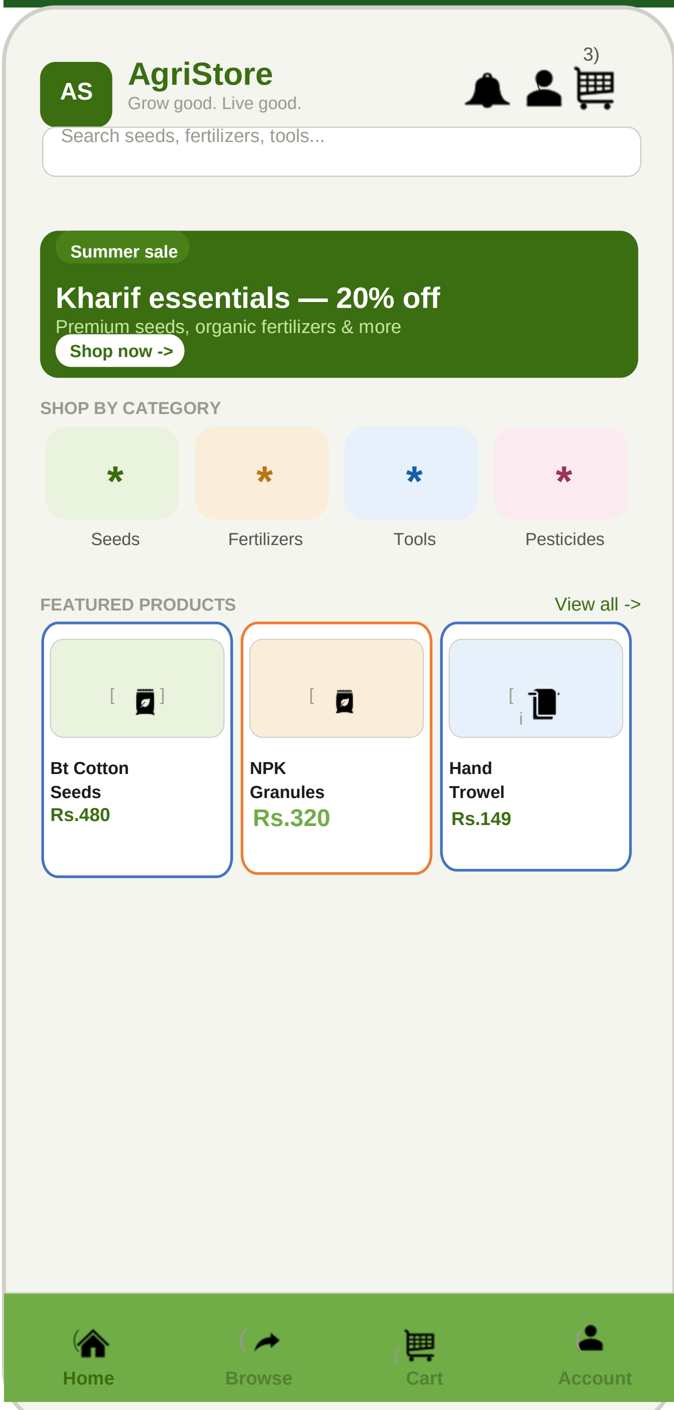
Req ID	Req Name	Req Description	Priority
NFR0101	Page Loading Time	Each page should load within 2 seconds on a 3G connection.	9
NFR0102	WCAG 2.1 Accessibility	The system must meet Web Content Accessibility Guidelines WCAG 2.1.	8
NFR0103	System Availability	System should maintain 99.5% uptime with scheduled maintenance windows.	9
NFR0104	Data Security	All user data must be encrypted at rest and in transit using SSL/TLS.	10
NFR0105	Scalability	System should handle 10,000 concurrent users without performance degradation.	8
NFR0106	Mobile Responsiveness	Application must be fully responsive on all screen sizes (mobile, tablet, desktop).	9

Question 34 – Wireframes & Prototypes

The following wireframes represent the key screens of the Online Agriculture Products Store. These were created to validate UI/UX with stakeholders before development begins.

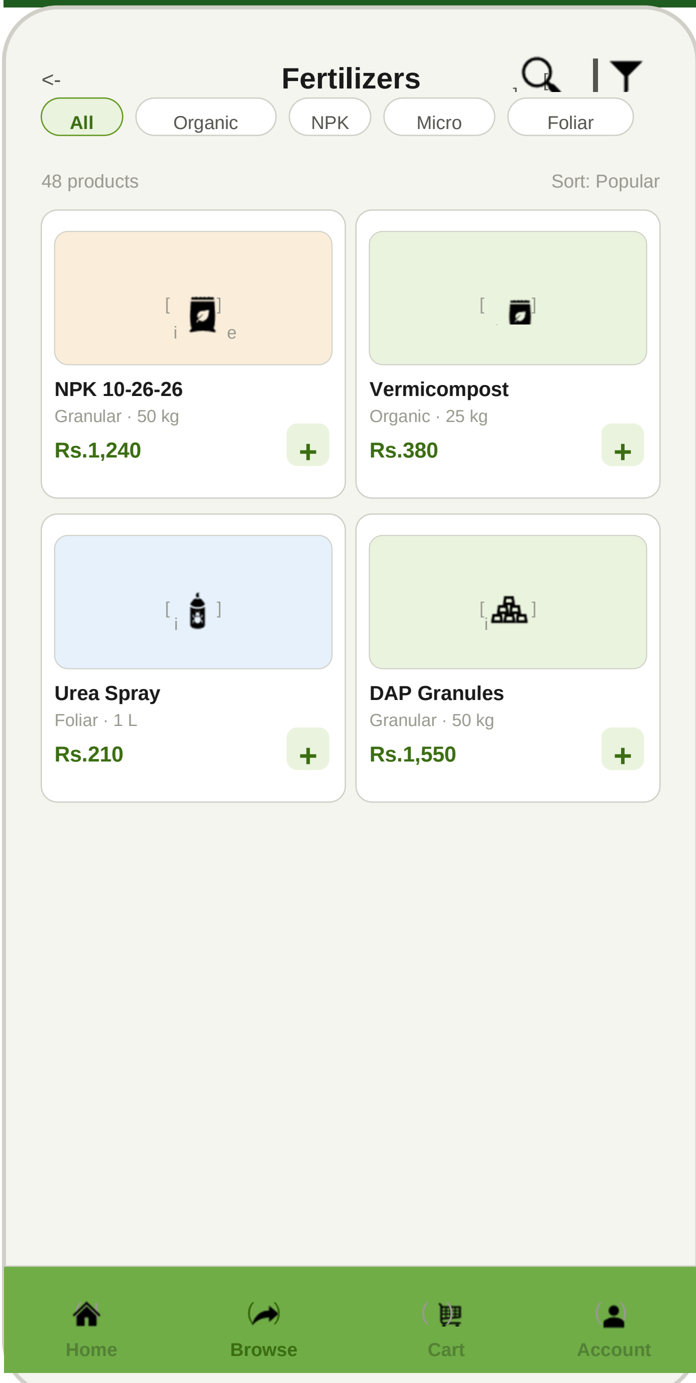
Wireframe 1: Home Page / Landing Page

[WIREFRAME: Wireframe 1: Home Page / Landing Page]



Wireframe 2: Category

Category



Wireframe 3: Product Detail

[WIREFRAME: Wireframe 3: Product Product Deail]

[image]

Certified organic

Bt Cotton Hybrid Seeds — 450 g

Rs.480 Rs.600 20% off

* * * * - 4.2 (318 reviews) In stock

High-yield Bt cotton hybrid. Bollworm resistant, good fibre quality. Suited for Kharif season.

Crop type	**Cotton**
Season	**Kharif**
Package	**450 g**
Coverage	**1 acre**

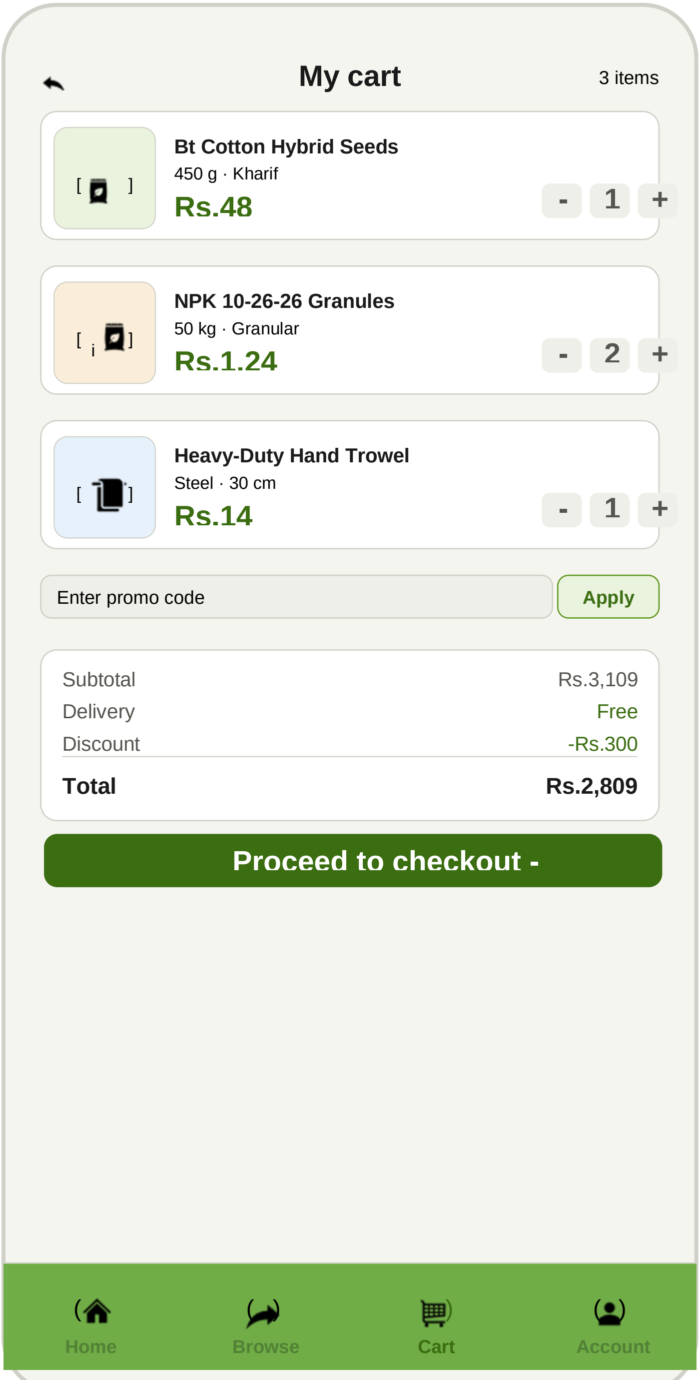
Qty: - 1 +

Add to cart — Rs.480

Home Browse Cart Account

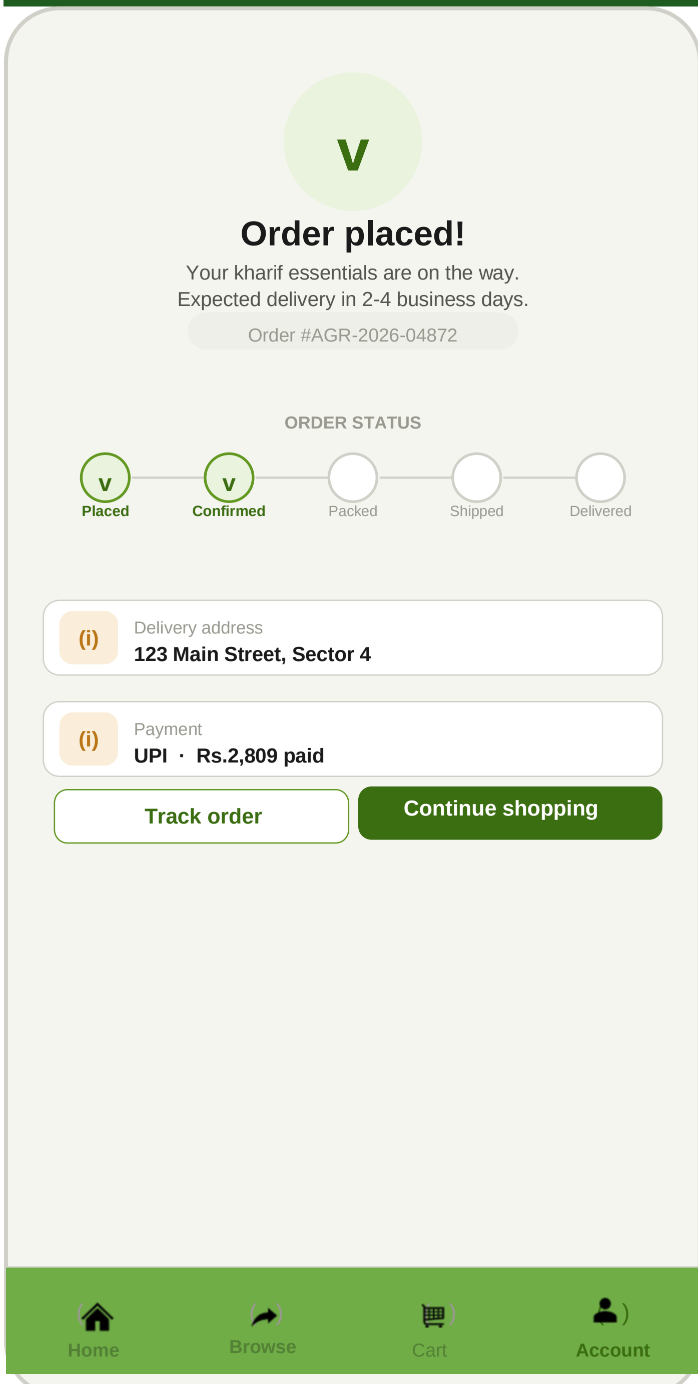
Wireframe 4: Shopping Cart

[WIREFRAME: Wireframe 4: Shopping Cart]



Wireframe 5: Order Confirmation

[WIREFRAME: Wireframe 5: Order confirmation]



Question 35 – Tools Used

Tool	Purpose	Used For
Balsamiq	Wireframing & Prototyping	Created screen for all 5 major pages: Home, Category, Product Details, Cart, Order confirmation.
Draw.io (diagrams.net)	Diagramming	Created Use Case Diagrams, Activity Diagrams, Data Flow Diagrams, ER Diagrams, and 3-Tier Architecture diagrams.
Microsoft Visio	Process Modeling & UML	Created detailed BPM flow charts, RACI Matrix, and technical architecture diagrams.
Microsoft Excel	RTM, Timesheets, Priority Tables	Maintained Requirements Traceability Matrix, BA Timesheets, and Requirements Priority tables.
Microsoft Word	Document Creation	Prepared BRD, FRS, Business Case, BA Approach Strategy, and all formal project documents.
JIRA	Project Management	Sprint planning, task tracking, defect management, and progress reporting.

Question 36 – Requirements Traceability Matrix (RTM)

The RTM ensures every requirement is tracked through design, development, testing, and UAT. This is used to respond to Mr. Henry and Peter's status queries.

Req ID	Req Name	Req Description	Design	D1	T1	D2	T2	D3	T3	D4	T4	UAT
FR0001	Farmer Registration	Farmers register on app	☑	☑	☑	-	-	-	-	-	-	☑
FR0002	Product Search	Search by category/filter	☑	-	-	☑	☑	-	-	-	-	☑
FR0005	Shopping Cart	Cart management	☑	-	-	☑	☑	-	-	-	-	☑
FR0006	Order Placement	Place order with address	☑	-	-	-	-	☑	☑	-	-	☑
FR0007	Payment Gateway	UPI/Card/COD payment	☑	-	-	-	-	☑	☑	-	-	☑
FR0009	Order Tracking	Real-time tracking	☑	-	-	-	-	-	-	☑	☑	☑
FR0012	Manufacturer Reg.	Manufacturer registration	☑	☑	☑	-	-	-	-	-	-	☑
FR0013	Product Upload	Manufacturer product upload	☑	-	-	☑	☑	-	-	-	-	☑
NFR0101	Page Load Time	Under 2 seconds	-	☑	☑	☑	☑	☑	☑	☑	☑	☑
NFR0104	Data Security	SSL/encryption	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑

Question 37 – Test Case Documents

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority
TC001	Farmer Registration – Valid Data	User is on Registration page	1. Enter valid Full Name, Mobile, Email, Password, Address. 2. Check T&C. 3. Click Register.	Registration successful. Welcome email sent. Redirect to dashboard.	High
TC002	Farmer Login – Valid Credentials	Farmer is registered	1. Enter valid email and password. 2. Click Login.	Login successful. Farmer dashboard displayed.	High
TC003	Product Search – Keyword Match	Farmer is logged in	1. Type 'Urea' in search box. 2. Click Search.	All Urea-related products displayed with name, price, image.	High
TC004	Add Product to Cart	Farmer is on product listing	1. Click Add to Cart on any product. 2. Navigate to Cart.	Product added to cart. Cart count incremented by 1.	High
TC005	Place Order – Successful Payment	Farmer has items in cart	1. Proceed to checkout. 2. Select UPI. 3. Complete payment. 4. Confirm order.	Order confirmed. Order ID generated. Confirmation email/SMS sent.	High
TC006	Order Tracking	Order has been placed	1. Go to My Orders. 2. Click Track Order on Order #001.	Order tracking page shows real-time status with timestamps.	High
TC007	Manufacturer Product Upload	Manufacturer is logged in	1. Navigate to Upload Product. 2. Fill all fields and upload images. 3. Click Publish.	Product published and visible in farmer's product catalog after admin approval.	High
TC008	Invalid Login – Wrong Password	User on Login page	1. Enter valid email but wrong password. 2. Click Login.	Error message displayed: 'Invalid credentials.	High

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority
				Please try again.' No dashboard access.	
TC009	Cart – Update Quantity	Farmer has product in cart	1. In cart, click '+' to increase Urea quantity from 1 to 3.	Cart updates. Total price recalculated correctly. Subtotal and GST updated.	Medium
TC010	Payment Failure Handling	Farmer is in payment step	1. Simulate payment gateway failure. 2. Click Pay.	Error displayed: 'Payment failed. Please try again.' Order not placed. Cart retained.	High

Question 38 – Database Schema & ER Diagram

Key Database Tables

Table Name	Key Columns	Description
USERS	user_id (PK), name, email, mobile, password_hash, role (FARMER/MANUFACTURER/ADMIN), status, created_at	Stores all registered users of the platform.
FARMER_PROFILE	farmer_id (PK), user_id (FK), village, district, state, pin_code, aadhaar_no	Farmer-specific profile details.
MANUFACTURER_PROFILE	manufacturer_id (PK), user_id (FK), company_name, gst_no, address, verified_status	Manufacturer company profile and verification status.
PRODUCTS	product_id (PK), manufacturer_id (FK), name, category, description, price, stock_qty, unit, expiry_date, status, created_at	Product catalog with all agricultural products.
PRODUCT_IMAGES	image_id (PK), product_id (FK), image_url, is_primary	Multiple product images linked to product.
ORDERS	order_id (PK), farmer_id (FK), total_amount, tax_amount, delivery_charges, delivery_address_id (FK), payment_status, order_status, placed_at	Master order record.
ORDER_ITEMS	item_id (PK), order_id (FK), product_id (FK), quantity, unit_price, subtotal	Line items for each order.
PAYMENTS	payment_id (PK), order_id (FK), payment_mode, transaction_id, amount, status, payment_at	Payment transaction records.
REVIEWS	review_id (PK), product_id (FK), farmer_id (FK), rating (1-5), comment, created_at	Product reviews and ratings by farmers.
ORDER_TRACKING	tracking_id (PK), order_id (FK), status, location, updated_at, notes	Order tracking history log.

Question 39 – Data Flow Diagram (DFD)

The Data Flow Diagram below shows the in-flow and out-flow of data when a Farmer places an order.

DFD – Farmer Order Placement (Level 1)

```
EXTERNAL ENTITIES:  [FARMER]   [MANUFACTURER]   [PAYMENT GATEWAY]   [LOGISTICS]

┌──────────────────────────────────────────────────────────────────────────────┐
│                     ONLINE AGRICULTURE PRODUCTS STORE SYSTEM                  │
│                                                                              │
│  [FARMER] ──── Login Credentials ────▶ (1.0 AUTHENTICATE USER)               │
│                                               │                              │
│                                       User Session Token                     │
│                                               ▼                              │
│  [FARMER] ─── Search Query/Category ─▶ (2.0 BROWSE PRODUCTS)                 │
│                                               │                              │
│                                       Product Data ◀── [PRODUCT DB]          │
│                                               │                              │
│                                       Product Results                        │
│                                               ▼                              │
│  [FARMER] ─── Add to Cart ──────────▶ (3.0 MANAGE CART)                      │
│                                               │                              │
│                                       Cart Data ──▶ [CART DB]                │
│                                               │                              │
│                                       Cart Summary                           │
│                                               ▼                              │
│  [FARMER] ─── Delivery Address ─────▶ (4.0 PLACE ORDER)                      │
│                                               │                              │
│                                       Order Data ──▶ [ORDER DB]              │
│                                               │                              │
│                                       Payment Request                        │
│                                               ▼                              │
│  [PAYMENT GATEWAY] ─ Payment Confirm ▶ (5.0 PROCESS PAYMENT)                 │
│                                               │                              │
│                                       Payment Record ──▶ [PAYMENT DB]        │
│                                               │                              │
│                                       Order Confirmation                     │
│                                               ▼                              │
│  [LOGISTICS] ─── Tracking Updates ──▶ (6.0 TRACK & DELIVER)                  │
│  [FARMER] ◀── Delivery Status / SMS ──────────────────────────               │
│  [MANUFACTURER] ◀── New Order Notification ───────────────────               │
│                                                                              │
└──────────────────────────────────────────────────────────────────────────────┘
```

Question 40 – Change Request Handling

Due to a change in Government Taxation Structure, a Change Request has been raised to update the Tax structure in the application.

Change Request Process

Step	Action	Responsible
1. Receive CR	Stakeholder raises a formal Change Request (CR Form) specifying what needs to change and why.	BA
2. CR Documentation	BA documents the CR with details: CR ID, title, description, requested by, date, justification.	BA
3. Impact Analysis	BA and PM assess impact on scope, timeline, budget, resources, and existing requirements.	BA, PM
4. Change Request Board (CRB)	CR presented to CRB (Mr. Karthik + Committee). Approved or rejected with documented reasons.	CRB
5. Update Documents	Upon approval, BA updates BRD, FRS, RTM, and test cases. Dev team informed.	BA
6. Implementation	Development team implements the tax structure change per updated requirements.	Dev Team
7. Testing & Validation	Testers validate the change against new requirements. BA performs review.	Testers, BA
8. Sign-off	Client signs off on the change. CR is closed and archived.	BA, Client

Question 41 – Change Request vs Enhancement

Scenario: Ben and Kevin Request Farmers to Sell Crop Yields with Auction System

As a Business Analyst, my response to Ben and Kevin's request:

Aspect	Change Request	Enhancement
Definition	A modification to an already agreed-upon requirement within the current project scope.	A new feature or capability that goes beyond the original project scope.
Impact	Affects existing functionality. May require redesign.	Adds entirely new functionality. Requires new development effort.
Process	Goes through formal Change Request Board (CRB) process.	Treated as a new phase or new project with its own budget and timeline.
Example	Changing tax calculation from 12% to 18% GST.	Adding ability for farmers to list and auction their crop yields.

BA Decision: This is an Enhancement

- The original scope of this project was defined as: Farmers buy agricultural products (seeds, fertilizers, pesticides) from Manufacturers.
- Allowing farmers to SELL their crop yields and introducing an Auction System is entirely outside the original scope.
- This represents new modules, new actor roles (Farmer as Seller, General Public as Buyer), new database design, and new workflows.
- Recommendation: Formally document this as an Enhancement Request. Raise it as a Phase 2 / New Project with separate budget, timeline, and scope definition.
- As BA, I will formally communicate this to Mr. Karthik and the Committee, and will NOT include this in the current project BRD without proper approval and scope revision.

Question 42 – Estimations (Man-Hours)

Phase	Resource	Hours/Week	Weeks	Total Hours
RG (Requirements Gathering)	PM, BA, 3 Stakeholders	8 hrs/day × 5 = 40	2	400
RA (Requirements Analysis)	BA, PM	40 hrs/week	4	320
Design	BA, Sr. Dev, DB Admin	40 hrs/week × 3	4	480
D1 (Development Sprint 1)	4 Java Devs, DB Admin	40 × 5	6	1,200
T1 (Testing Sprint 1)	2 Testers, BA	40 × 3	4	480
D2 (Development Sprint 2)	4 Java Devs, DB Admin	40 × 5	6	1,200
T2 (Testing Sprint 2)	2 Testers, BA	40 × 3	4	480
D3 (Development Sprint 3)	4 Java Devs, NW Admin	40 × 5	6	1,200
T3 (Testing Sprint 3)	2 Testers, BA	40 × 3	4	480
D4 (Development Sprint 4)	4 Java Devs, DB Admin	40 × 5	4	800
T4 (Testing Sprint 4)	2 Testers, BA	40 × 3	4	480
UAT	BA, PM, 3 Stakeholders	40 × 5	6	1,200
Deployment & Implementation	All Resources	40 × 10	2	800
TOTAL	All Resources	—	~50 weeks	~9,520 hrs

Note: Total estimated effort = ~9,520 man-hours across all phases. With 10 resources at an average blended rate, this fits within the INR 2 Crore budget.

Question 43 – UAT Acceptance Process

UAT Process Steps

- UAT Planning: BA (Mrs Neha) prepares UAT Plan including scope, objectives, schedule, participants (Peter, Kevin, Ben), and sign-off criteria.
- UAT Environment Setup: DB Admin (Mr. John) and Dev team set up a UAT server environment with production-like data.
- UAT Test Scripts: BA prepares test scripts based on Business Requirements and hands to Peter, Kevin, Ben for execution.
- UAT Execution: Farmers (Peter, Kevin, Ben) test all major workflows: Registration, Search, Cart, Order, Payment, Tracking.
- Defect Logging: Any issues found are logged in JIRA as UAT defects and triaged by BA and PM.
- Defect Fixes: Development team fixes defects. BA re-tests to validate fixes.
- UAT Sign-Off: Upon successful completion with no critical defects, BA prepares Client Project Acceptance Form (CPAF).
- CPAF Signed: Mr. Henry / Committee signs CPAF. This formally closes the UAT phase.

Project Closure Activities

- Final deliverables handed over to SOONY Committee
- Knowledge Transfer sessions conducted for any SOONY technical staff
- All project documents archived in repository
- Project Closure Document prepared by PM and BA
- Lessons Learned document compiled
- Team recognition and project sign-off meeting held

Question 44 – Project Closure Document

Section	Details
Project Name	Online Agriculture Products Store
Project Sponsor	Mr. Henry / SOONY Company
Development Company	APT IT Solutions
Start Date	Month 1 of Project
End Date	Month 18 of Project
Original Budget	INR 2 Crores
Final Budget Used	Within approved budget
Project Manager	Mr. Vandanam
Business Analyst	Mrs. Neha Bhatt (BA)
Objectives Achieved	1. Farmers can browse and purchase agricultural products online. 2. Manufacturers can upload and manage product listings. 3. Admin can manage the platform. 4. Secure payment and real-time tracking implemented. 5. Application deployed and live.
Deliverables Completed	BRD, FRS, Use Case Diagrams, Activity Diagrams, Wireframes, RTM, Test Cases, UAT, DFD, ER Diagram, Gantt Chart, Deployed Application.
Outstanding Issues / Risks	Phase 2 enhancement (Farmer Crop Selling & Auction – pending separate approval).
Lessons Learned	Early wireframing greatly helped stakeholder alignment. Quarterly audits kept project on track. V-Model's structured testing reduced late-stage defects significantly.
Client Acceptance	CPAF signed by Mr. Henry and Committee upon successful UAT.
Project Status	☑ CLOSED – Successfully Delivered

— End of Capstone Project Document —